

A25 Team: OpenSilicon

Design of 10-Bit SAR ADC for IoT Applications

Discord name	Affiliation	Experience	Role
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Introduction

Project Goal: The goal is to design and verify a 10-bit charge-redistribution SAR ADC using the GF180MCU process. The design targets low-power IoT and sensor-interface applications at sampling rates up to 1 MS/s.

Design: The ADC will use a differential capacitive DAC, comparator, and synthesizable SAR control logic. Initial work will focus on behavioral modeling, error budgeting, and block-level schematic verification before proceeding to layout and integration.

Application: Low-power, medium-resolution data acquisition for IoT sensors, battery monitoring, and industrial sensing systems.

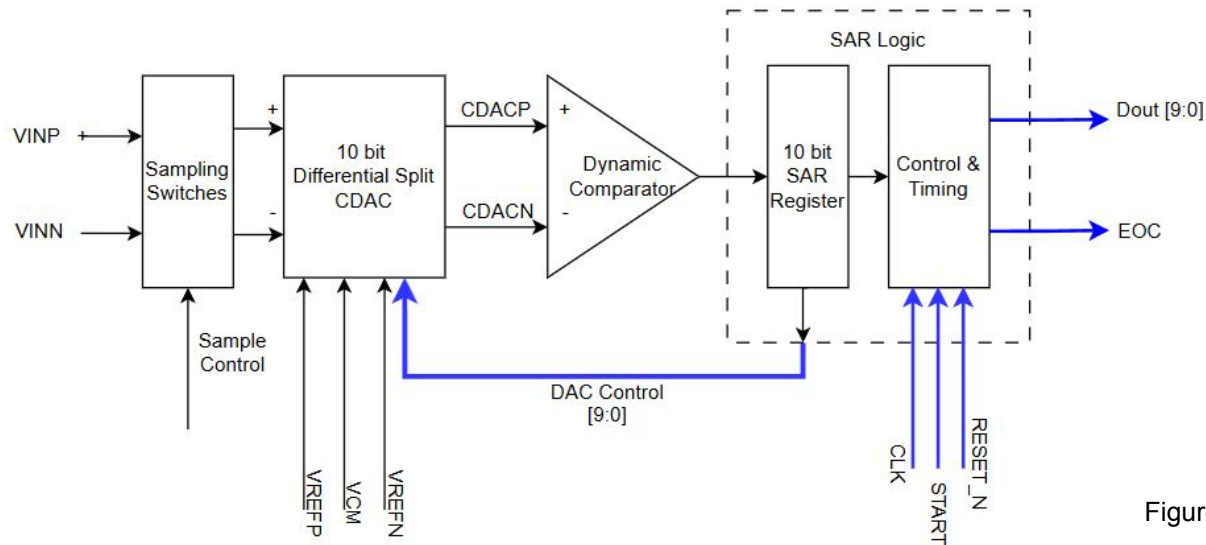


Figure 1, 10-Bit SAR ADC block diagram

Targeted Specs

Parameter	Guaranteed (safe)	Target (stretch)	Notes
Resolution	10 bits	10 bits	
Sample rate	500 kS/s	1 MS/s	
Supply	3.3 V	3.3 V	
ENOB	≥ 8.5 bits	≥ 9.2 bits	SKY130 10-bit reference hit 9.31 ENOB
SNDR	≥ 53 dB	≥ 57 dB	$\text{SNDR} = 6.02 \times \text{ENOB} + 1.76$
SFDR	≥ 65 dBc	≥ 72 dBc	SKY130 10-bit hit 72.2 dB
INL / DNL	$\leq \pm 1$ LSB / ± 1 LSB	$\leq \pm 0.5$ LSB	
Power	≤ 5 mW	≤ 2 mW	Published 0.18 μm 10-bit designs: 0.43 μW – 5.23 mW range
FOM (Walden)	≤ 20 fJ/step	≤ 10 fJ/step	6.38 fJ achieved in 0.18 μm at 100 kS/s
Input range	0 – 3.0 V	0 – 3.0 V	
Area	≤ 0.4 mm ²	≤ 0.25 mm ²	SKY130 12-bit open-source: 0.175 mm

Pad type	Count	Cell name
Analog inputs (VINP, VINN, VREFP, VCM)	4	gf180mcu_fd_io__asig_5p0
Digital inputs (CLK, START, RESET_N)	3	gf180mcu_fd_io__in_c
Digital outputs DOUT[9:0] + EOC	11	gf180mcu_fd_io__bi_t
Power/ground pads	2	gf180mcu_fd_io__dvdd / dvss
Total	20	

References

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- B. Razavi, “The StrongARM Latch,” *IEEE Solid-State Circuits Magazine*, vol. 7, no. 2, pp. 12–17, Spring 2015, doi: 10.1109/MSSC.2015.2418155.